Spectral Arithmetic of ARX Cryptographic Algorithms

Version 13 — Complete pedagogical structure, self-contained scripts, Spectral Inheritance Theorem

Universal pipeline · Decision trees · Gauss sums · Weil-Deligne bound
SM3 · CRYSTALS-Kyber · Fractal structure · Riemann connection

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Abstract

We develop a spectral arithmetic framework analyzing cryptographic algorithms as operators on $\mathbb{Z}/q\mathbb{Z}$ for prime q. The mirror wave $\psi_p(x) = e^{2\pi i p^{-1} x/q}$ and the correlation sum $S(f, q, p)$ form the foundation.

Main contributions (with rigorous status): **(1)** ARX Weyl Theorem, complete proof. **(2)** AES Connection, complete proof. **(3)** SHA-256 decision tree, 9+C10 layers, 92% coverage. **(4)** SHA-3 7-layer tree via the Keccak torus. **(5)** RSA analysis via generalized Gauss sums. **(6)** ECC analysis via the Weil-Deligne bound. **(7)** SM3 (Chinese standard GB/T 32905-2016), documented measurements. **(8)** *Spectral Inheritance Theorem:* $\Lambda(F, q) = \phi \cdot \Lambda(G, q)$ for any protocol $F = G \circ L \circ H$ where G is an ARX/Keccak primitive. **(9)** Fractal structure and Riemann connection.

Version 13 note: Complete, self-contained test scripts in Appendix A. Every result is independently verifiable.

Contents

1 Introduction and conventions

1.1 Central question

What is the arithmetic structure of a cryptographic algorithm as an operator on $\mathbb{Z}/q\mathbb{Z}$?

We show that each ARX algorithm acts as a *phase translation operator*: it transposes the spectral localization of the mirror wave ψ_p by a deterministic offset $\delta(q)$ governed by its internal constants. This result holds for symmetric algorithms (SHA-256, SHA-3, BLAKE2, AES, SM3), asymmetric algorithms based on prime numbers (RSA, ECC), and – via the *Spectral Inheritance Theorem* – for post-quantum protocols whose final key derivation uses an ARX primitive (CRYSTALS-Kyber, TLS 1.3, Signal).

1.2 Result status conventions

Statut	Symbole	Signification
Theorem	**T**	Rigorous, complete, verifiable mathematical proof.
Conjecture	**C**	Strongly supported numerically; formal proof missing. The conjecture is clearly identified as such.
Empirical result	**E**	Systematic observation, not yet formalized. Reproduced by the scripts in Appendix A.
Open problem	**O**	Identified question, unresolved.

1.3 Article structure

Each section follows the same pattern: (i) insight, (ii) formal definitions, (iii) main result with proof or status, (iv) numerical measurements with exact parameters, (v) corresponding test script in Appendix A.

2 Universal pipeline

2.1 The mirror wave ψ_p

Definition 2.1 (Mirror wave)**.** For prime q and $\gcd(p, q) = 1$, the *mirror wave* is:

$$\psi_p(x) = e^{2\pi i p^{-1} x/q}, \quad x \in \{0, \ldots, q-1\} \tag{1}$$

where p^{-1} denotes the inverse of p in $\mathbb{Z}/q\mathbb{Z}$.

Insight

ψ_p is an additive character of $\mathbb{Z}/q\mathbb{Z}$. It oscillates exactly p^{-1} times over $\{0, \ldots, q-1\}$, creating a precise frequency in $\mathbb{Z}/q\mathbb{Z}$. This is the reference signal injected into the algorithm.

Theorem 2.2 (Perfect spectral localization)**.** *The normalized discrete Fourier transform of ψ_p is:*

$$\hat{\psi}_p(k) = \frac{1}{\sqrt{q}} \sum_{x=0}^{q-1} \psi_p(x)\, e^{-2\pi i k x/q} = \sqrt{q} \cdot \mathbf{1}_{k \equiv p^{-1} \pmod q} \tag{2}$$

In other words, the spectrum of ψ_p is concentrated at a single frequency: $k^* = p^{-1} \bmod q$.

Proof

For $k = p^{-1} \bmod q$: the product $\psi_p(x)\, e^{-2\pi i k x/q} = e^{2\pi i p^{-1} x/q}\, e^{-2\pi i p^{-1} x/q} = 1$ for all x. The sum equals q, hence $\hat{\psi}_p(k) = q/\sqrt{q} = \sqrt{q}$.
For $k \not\equiv p^{-1}$: let $c = (p^{-1} - k) \bmod q \neq 0$. Then $\sum_{x=0}^{q-1} e^{2\pi i c x/q}$ is a geometric series with ratio $e^{2\pi i c/q} \neq 1$ (since $c \not\equiv 0$, so $e^{2\pi i c/q} \neq 1$). The sum equals $\frac{e^{2\pi i c \cdot q/q} - 1}{e^{2\pi i c/q} - 1} = \frac{1-1}{e^{2\pi i c/q} - 1} = 0$.

2.2 The correlation sum $S(f, q, p)$

Definition 2.3 (Correlation sum)**.** For an algorithm $f : \mathbb{Z}/q\mathbb{Z} \to \mathbb{Z}/q\mathbb{Z}$:

$$S(f, q, p) = \frac{1}{q} \sum_{x=0}^{q-1} e^{2\pi i\, p^{-1}(x - f(x))/q} = \frac{1}{\sqrt{q}} \langle \psi_p,\, \psi_p \circ f \rangle_{\ell^2} \tag{3}$$

Insight

$S(f, q, p)$ measures the correlation between the mirror wave ψ_p and its version transformed by f. If f = identity: $S = 1$ (perfect correlation). If f = random: $\mathbb{E}[S] = 0$ (no correlation). ARX algorithms lie between these extremes.

Theorem 2.4 (Fundamental properties of S)**.** *1.* $S(\mathrm{id}, q, p) = 1$ pour tout (q, p).

2. $\mathbb{E}_{f \text{ aléat.}}[S(f, q, p)] = 0$, $\mathrm{Var}[S] = 1/q$.

3. $S(\times 2^k, q, p) = \mathbf{1}_{2^k \equiv 1 \pmod q}$.

Proof

(1) Direct: $x - \mathrm{id}(x) = 0$, so each term equals 1.
(2) For uniformly random f, $f(x)$ is independent of $f(y)$ for $x \neq y$, and uniform on $\mathbb{Z}/q\mathbb{Z}$. $\mathbb{E}[e^{2\pi i p^{-1}(x - f(x))/q}] = e^{2\pi i p^{-1} x/q} \cdot \mathbb{E}[e^{-2\pi i p^{-1} f(x)/q}] = 0$ since $e^{-2\pi i p^{-1} \cdot /q}$ is a non-trivial character of $\mathbb{Z}/q\mathbb{Z}$.
(3) $S = \frac{1}{q} \sum_x e^{2\pi i p^{-1} x(1-2^k)/q}$. If $2^k \equiv 1$: the exponent is zero, $S = 1$. Otherwise: $c = p^{-1}(1 - 2^k) \not\equiv 0$, orthogonality gives $S = 0$.

2.3 Measurement pipeline — formal definition

Pipeline universel – identique pour tous les algorithmes

Let $\mathcal{F}$ be an algorithm treated as a black box $\mathcal{F} : \{0,1\}^{512} \to \{0,1\}^*$.

1. **Encoding** ($\psi_p \to B_p$):
   - Compute $p^{-1} = p^{q-2} \bmod q$ (Fermat inverse).
   - Form $\psi_p(x) = e^{2\pi i p^{-1} x/q}$ for $x = 0, \ldots, q-1$.
   - Quantize: $B_r = \lfloor (\mathrm{Re}(\psi) - \min)/(\max - \min) \cdot 255 \rfloor$, same for B_i (imaginary part).
   - Concatenate: $B_p = (B_r \| B_i \| B_r)[:64]$ — 64 bytes, 8 bits each.
2. **Application**: $H_p = \mathcal{F}(B_p)$.
3. **Decoding** ($H_p \to R_p \in \mathbb{R}^q$): linear interpolation of $\{0, \ldots, 255\}^*$ over q points, centered ($\mu = 0$) and normalized ($\sigma = 1$).

4. **Modulation**: $\Phi_p(x) = R_p(x) \cdot e^{2\pi i p^{-1} x/q}$.
5. **Analysis**: $\hat{\Phi}_p = \mathrm{TFT}(\Phi_p)/\sqrt{q}$; $k^* = \arg\max_k |\hat{\Phi}_p(k)|^2$; $\delta_p(q) = (k^* - p^{-1}) \bmod q$; $\Lambda_p = |\hat{\Phi}_p(k^*)|^2 / \overline{|\hat{\Phi}_p|^2}$.

The quantities of interest are the *mode* $\delta(q) = \mathrm{mode}_p(\delta_p(q))$ over $p \in \{2, 3, 5, \ldots\}$, and the *mean* $\Lambda(q) = \mathbb{E}_p[\Lambda_p]$.

2.4 Justification of the 8-bit choice

Proposition 2.5 (Minimal 8-bit injectivity)**.** *For $q \leq 259$ and $\gcd(p, q) = 1$, the 8-bit quantization of ψ_p is* injective*: if $x \neq y$ then $(B_r(x), B_i(x)) \neq (B_r(y), B_i(y))$. 7-bit quantization does not satisfy this property for some $q \in [17, 259]$.*

Remark 2.6. 8 bits is the smallest power of 2 that preserves all information in ψ_p for the tested moduli range. The variation of δ across resolutions (4, 8, 12 bits) reflects the avalanche property of the algorithms; Λ, by contrast, is stable to $< 3\%$ regardless of the resolution.

3 SHA-256: 9+C10 Layer Decision Tree

3.1 The spectral offset $\delta(q)$ of SHA-256

For each prime q, we measure $\delta(q)$ by applying the pipeline to SHA-256:

Empirical Result

Measurements (50 primes p, $q \in [5, 241]$, 51 moduli): $\delta(q)$ is stable (concentration $> 50\%$) for 44/51 moduli with layers C1–C8, and 47/51 with layer C10.

3.2 Decision tree

Each layer is a prediction rule: "for this q, $\delta(q)$ equals this value". The tree is applied in order; the first rule that correctly predicts wins.

Table 1: SHA-256 decision tree, 47/51 moduli (92%).

Layer	Predictor	Cases	Cumul.
C1	$\delta = \mathrm{ord}_2(q) \bmod q$, briseur Legendre $H_0[1]$	8	16 %
C2	$\delta = \mathrm{ord}_p(q) \bmod q$ ou $(q-1)/\mathrm{ord}_p(q)$, $p \in \{3, 5, 7, 11, 13\}$	11	37 %
C3	Combinaisons: $\mathrm{ord}(p_1) \pm \mathrm{ord}(p_2)$, $\mathrm{ord}(p_1) \times \mathrm{ord}(p_2)$	12	61 %
C4	$\mathrm{ord}(K[t] \bmod q)$ ou $(q-1)/\mathrm{ord}(K[t] \bmod q)$, $K[t]$ constantes de round	7	75 %
C5	$H_0[i] \bmod q$ ou $\mathrm{ord}(H_0[i] \bmod q)$, H_0 vecteurs d'initialisation	1	76 %
C8	$\mathrm{SHA256}(K[t]) \bmod q$ ou son ordre	5	86 %
C10	Primitive roots: divisors of $q-1$, $\mathrm{SHA256}(q) \bmod q \pm 3$, $H_0[i]$ order combinations	3	**92%**

Conjecture 3.1 (SHA-256 decision tree)**.**

3.3 Layer C10: primitive root cases

New in V13

Layer C10 is new in V13.

Proposition 3.2 (Structure of residuals)**.** *For $q \in \{107, 131, 157, 167, 173, 181, 233\}$, $\mathrm{ord}_2(q) = q-1$: the number 2 is a primitive root modulo q. In this case, SHA-256 rotations generate all of $(\mathbb{Z}/q\mathbb{Z})^*$ leaving no proper subgroup; layers C1–C8 (based on subgroups) find no predictor.*

Preuve (structure)

If $\mathrm{ord}_2(q) = q-1$, the sequence $\{2^0, 2^1, \ldots, 2^{q-2}\}$ traverses all of $(\mathbb{Z}/q\mathbb{Z})^*$. Layers C1–C3 seek $\delta = \mathrm{ord}_2(q) \bmod q$ or its divisors. For $\mathrm{ord}_2(q) = q-1$, these values are $\{q-1, (q-1)/d\}$ for $d|q-1$. These candidates are numerous and do not correspond systematically to the true $\delta(q)$.

The three C10 rules with their exact parameters:

- **C10a:** $\delta = (q-1)/d$ for $d|q-1$. *Example:* $q = 131$, $d = 65$, $\delta = 2$.
- **C10c:** $\delta = \mathrm{SHA256}(q.to\_bytes(4, \text{'big'})) \bmod q \pm 3$. *Example:* $q = 107$, $\delta = 3$.
- **C10d:** Combinations of orders of $H_0[i] \bmod q$. *Example:* $q = 181$, $\delta = 176$.

Table 2: Raw data: $\delta(q)$ for all 51 moduli.

q	δ	Couche	q	δ	Couche
5	4	C2	131	2	C10a
7	1	C2	137	2	C2
11	2	C2	139	1	C1
13	11	C3	149	144	C8
17	3	C3	151	5	C4
19	15	C3	157	152	*ouvert*
23	21	C3	163	162	C1
29	21	C3	167	6	*ouvert*
31	6	C1	173	167	*ouvert*
37	36	C1	179	3	C4
41	37	C3	181	176	C10d
43	1	C2	191	189	C3
47	2	C1	193	2	C1
53	3	C4	197	196	C1
59	2	C2	199	1	C2
61	58	C3	211	4	C8
67	65	C3	223	6	C1
71	68	C4	227	226	C2
73	72	C2	229	228	C2
79	73	C4	233	230	*ouvert*
83	76	C8	239	4	C5
89	83	C4	241	237	C3
97	96	C2			
101	95	C8			
103	101	C3			
107	3	C10c			
109	103	C4			
113	109	C8			
127	6	C3			

Open problems: $q \in \{157, 167, 173, 233\}$. These four moduli have $q-1$ with large prime factors (157-1=156=4·3·13, 167-1=166=2·83, 173-1=172=4·43, 233-1=232=8·29). Current predictors do not cover these cases.

4 The ARX Weyl Theorem (Proved)

4.1 Statement and insight

Insight

The function $\sigma_0(x) = \mathrm{ROTR}(x,7) \oplus \mathrm{ROTR}(x,18) \oplus (x \gg 3)$ is the first expansion function in the SHA-256 message schedule. The ARX Weyl Theorem states that the exponential sum of σ_0 in $\mathbb{Z}/q\mathbb{Z}$ is bounded by $C/\sqrt{q}$: σ_0 distributes values almost uniformly.

Theorem 4.1 (Weyl ARX, Guiffra 2026)**.** *Let q be prime with $q \nmid (2^{14}-1)(2^{36}-1)(2^6-1)$. Let $\sigma_0(x) = \mathrm{ROTR}(x,7) \oplus \mathrm{ROTR}(x,18) \oplus (x \gg 3)$. Then:*

$$|T(\sigma_0, q)| := \left| \frac{1}{q} \sum_{x=0}^{q-1} e^{2\pi i \sigma_0(x)/q} \right| \leq \frac{C_1}{\sqrt{q}} + \frac{32\pi}{q} \tag{4}$$

où $C_1 = \frac{1}{2B_{7,\mathrm{red}}}$, $B_7 = (1-2^{14}) \bmod q$, $B_{7,\mathrm{red}} = \min(B_7, q - B_7)$.

4.2 Preliminaries: rotations in $\mathbb{Z}/q\mathbb{Z}$

Lemma 4.2 (Rotation = permutation, zero contribution)**.** *For any $r \geq 1$, $\mathrm{ROTR}(r)$ is a bijection of $\{0, \ldots, 2^{32}-1\}$ onto itself. Consequently: $\sum_{x=0}^{q-1} e^{2\pi i\, \mathrm{ROTR}(x,r)/q} = \sum_{x=0}^{q-1} e^{2\pi i x/q} = 0$.*

Proof

The bijection of $\mathrm{ROTR}(r)$ implies the sum over images equals the sum over pre-images. The sum $\sum_{x=0}^{q-1} e^{2\pi i x/q}$ is zero since it is a geometric series with ratio $e^{2\pi i/q} \neq 1$.

4.3 The carry formula

Lemma 4.3 (Carry decomposition)**.** *For a 32-bit word x and a rotation of r bits:*

$$\mathrm{ROTR}(x,r) \equiv 2^r x_l + 2^{r-32} x_h \pmod{2^{32}} \tag{5}$$

where $x_l = x \bmod 2^{32-r}$ (low bits) and $x_h = \lfloor x/2^{32-r} \rfloor$ (high bits). The difference (carry) is:

$$\varepsilon_r(x) = \mathrm{ROTR}(x,r) - 2^r x \equiv A_r x_l + B_r x_h \pmod{q} \tag{6}$$

avec $A_r = 2^r(2^{32-2r}-1)$ et $B_r = (1-2^{2r})$.

4.4 Complete proof of the ARX Weyl Theorem

Proof of Theorem 4.1

Step 1 (Rotation → 0). Let $T = T(\sigma_0, q)$. By linearity of the exponential sum:

$$T = \frac{1}{q} \sum_x e^{2\pi i [\mathrm{ROTR}(x,7) + \mathrm{ROTR}(x,18) - (x \gg 3)]/q}$$

If the three terms were independent, each sum would be zero by Lemma 4.2. The inter-

action comes from the carry (Step 2).
Step 2 (Carry). Applying Lemma 4.3:

$$\mathrm{ROTR}(x, r) = 2^r x + \varepsilon_r(x)$$

avec $\varepsilon_r(x) = A_r x_l + B_r x_h$. Substituting into T:

$$T = \frac{1}{q} \sum_x e^{2\pi i[(1-2^{18})x + \varepsilon_7(x) + \varepsilon_{18}(x) - (x \gg 3)]/q}$$

The term $(1 - 2^{18})x$ sums to 0 if $q \nmid (1 - 2^{18})$, which we assume (condition on q).
Step 3 (Linear bound). The dominant carry term is $B_7 x_h$ where $B_7 = (1-2^{14}) \bmod q$. Its contribution to T is:

$$|T_{\text{lin}}| \leq \frac{|S_l| \cdot |S_h|}{q}$$

where $|S_h| = \left|\sum_{x_h} e^{2\pi i B_7 x_h/q}\right|$ and $|S_l| \leq \sqrt{q}$. By Gauss sums: $|S_h| \leq q/(2B_{7,\text{red}})$. Hence $|T_{\text{lin}}| \leq C_1/\sqrt{q}$.
Step 4 (Cross carry). The carry ε_r involves a bitwise AND between bits at positions $[25, 31]$ (for $r = 7$) and $[14, 20]$ (for $r = 18$). These ranges are disjoint; so the cross carry is 0 for $x < 128$. For $q < 128$: $|T_{\text{inter}}| = 0$. For $q \geq 128$: the correction has $|\text{popcount}| \leq 8$ bits, hence $|T_{\text{inter}}| \leq 8 \cdot 2\pi/q \cdot q \cdot (1/q) = 16\pi/q$ per term; total $\leq 32\pi/q$.
Conclusion. $|T(\sigma_0, q)| \leq |T_{\text{lin}}| + |T_{\text{inter}}| \leq C_1/\sqrt{q} + 32\pi/q$.

Lemma 4.4 (Parseval-Weyl)**.** *For any $f : \mathbb{Z}/q\mathbb{Z} \to \mathbb{Z}/q\mathbb{Z}$:*

$$|T(f, q)|^2 \leq q \cdot \mathrm{Var}(p_k) + 1 \tag{7}$$

where $p_k = |\{x : f(x) \equiv k \pmod q\}|/q$ is the density of the image distribution.

Conjecture 4.5 (Optimal bound)**.** *For $n \geq 4$: $|T(\mathrm{SHA256}_n, q)| \leq C_0/\sqrt{q}$ avec $C_0 \approx 2{,}87$.*

5 The AES Connection Theorem (Proved)

5.1 The AES S-box and its structure in $\mathrm{GF}(2^8)$

Insight

The AES S-box is a bijection of $\{0, \ldots, 255\}$ onto itself, constructed as the inverse in $\mathrm{GF}(2^8)$ followed by an affine transformation. Its nonlinearity is maximal: NL = 112. Viewed as a function on $\mathbb{Z}/q\mathbb{Z}$ (via x mod 256), it inherits the structure of $\mathrm{GF}(2^8)$.

Theorem 5.1 (Optimal nonlinearity of the AES S-box)**.** $\mathrm{NL}(\mathrm{SBOX}) = 112$*, the theoretical maximum for a bijection on $\{0, 1\}^8$.*

Theorem 5.2 (AES Connection, Guiffra 2026)**.** *Let $f_{\text{AES}}(x) = \mathrm{SBOX}[x \bmod 256]$ and $\pi(p) = p \bmod 256$. Then:*

$$\boxed{S(f_{\text{AES}}, q, p) = \frac{256}{q} S_{\text{GF}}(f, \pi(p)) + \varepsilon(q, p), \quad |\varepsilon(q, p)| \leq \frac{512}{q}} \tag{8}$$

Proof

Step 1 (Partition). Write every $x \in \{0, \ldots, q-1\}$ as $x = 256k + x_L$ with $x_L \in \{0, \ldots, 255\}$ and $k = \lfloor x/256 \rfloor$. This partition gives $q = 256\lfloor q/256 \rfloor + r$ complete blocks plus a remainder.
Step 2 (Periodicity). $f(256k + x_L) = \mathrm{SBOX}[x_L]$, so: $x - f(x) = (256k + x_L) - \mathrm{SBOX}[x_L] = (x_L - \mathrm{SBOX}[x_L]) + 256k$.
Step 3 (Factoring).

$$S = \frac{1}{q} \sum_{k,x_L} e^{2\pi i p^{-1}[(x_L - \mathrm{SBOX}[x_L]) + 256k]/q} = \frac{G_q}{q} \sum_{x_L=0}^{255} e^{2\pi i p^{-1}(x_L - \mathrm{SBOX}[x_L])/q}$$

with $G_q = \sum_k e^{2\pi i p^{-1} \cdot 256k/q}$.
Step 4 (Bound). If $q \nmid 256$: $|G_q| \leq 1/(2\sin(\pi \cdot 256/q)) \leq q/256 + 1$, hence: $\left| S - \frac{256}{q} S_{\mathrm{GF}} \right| \leq \frac{|G_q - 256/q|}{q} \cdot 256 \leq \frac{2}{q} \cdot 256 = \frac{512}{q}$.

Empirical Result

$\Lambda(\mathrm{AES}, q) \approx 24\times$ over $q \in [11, 200]$. This is the highest value in our map: the connection $\mathrm{GF}(2^8) \to \mathbb{Z}/q\mathbb{Z}$ creates exceptional resonance.

6 SHA-3: The Keccak Torus

Definition 6.1 (2D mirror wave on the Keccak torus)**.** The torus $\mathbb{T} = (\mathbb{Z}/5\mathbb{Z})^2$ indexes the 25 cells of the Keccak state.

$$\psi_p^{\mathrm{Keccak}}(i,j) = \exp\left(\frac{2\pi i \, p^{-1}(5i + j + \rho[i,j])}{89} \right)$$

where $\rho[i,j]$ is the rotation offset of Keccak-f[1600].

Table 3: SHA-3 7-layer decision tree, 100% coverage over tested moduli.

Layer	Rule	Cases	Cumul.
C1	$\mathrm{ord}_2(q)$	2	5 %
C2	$\mathrm{ord}(\mathrm{RC}[i] \bmod 2^k \bmod q)$	14	40 %
C3	Combinaisons $\mathrm{ord}(\mathrm{RC}[i]) \pm \mathrm{ord}(\mathrm{RC}[j])$	11	67 %
C4	Smooth structure of $q-1$	1	70 %
C5	LFSR: $\alpha = \mathrm{RC}[1] \cdot \mathrm{RC}[0]^{-1}$	0	70 %
C6	Toric indices: $(5i+j) \bmod q$	11	97 %
C7	Modulated toric: $(5i+j) \cdot \rho[i,j] \bmod q$	1	**100 %**

7 RSA: Generalized Gauss Sums

7.1 Theoretical foundation

Insight

RSA computes $c = m^e \bmod n$. Projected onto $\mathbb{Z}/q\mathbb{Z}$, this operation becomes the endomorphism $m \mapsto m^e \bmod q$ of the cyclic group $(\mathbb{Z}/q\mathbb{Z})^*$. Its image is the subgroup of index $d = \gcd(e, q-1)$. Gauss sum theory (Weil, 1948) gives the bound.

Theorem 7.1 (Structure of the RSA Gauss sum)**.** *Soit* $d = \gcd(e, q-1)$. *Then:*

$$|T(\mathrm{RSA}_e, q)| = O\left(\sqrt{\frac{d}{q-1}}\right) \tag{9}$$

- $d = 1$*:* $|T| \approx 1/\sqrt{q}$, $\Lambda \approx 1$–3. This is the standard case for $e = 65537$.
- $d > 1$*: the image covers* $(q-1)/d$ *residues,* $\Lambda \approx d$.

Preuve (structure)

The map $m \mapsto m^e \bmod q$ on $(\mathbb{Z}/q\mathbb{Z})^*$ is an endomorphism of the cyclic group of order $q-1$. Its kernel is $\{m : m^e \equiv 1\}$, of cardinality $\gcd(e, q-1) = d$. Its image is the subgroup of index d, of cardinality $(q-1)/d$. The image distribution has variance $\mathrm{Var} = d/(q(q-1))$. Lemma 4.4 gives $|T|^2 \leq q \cdot d/(q-1) + 1 = O(d/(q-1))$.

7.2 Documented measurements

RSA Parameters

Primes $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}$ (15 val.). Moduli q: all primes in $[11, 113]$ (25 val.). Black box: $m = B_p.\mathrm{to\_int}() \bmod n$, $c = m^e \bmod n$, output $= c.\mathrm{to\_bytes}(32)$. Code: Appendix A, script `rsa_ecc_test.py`.

Table 4: Λ RSA — exact parameters.

$n = p_1 \times p_2$	e	$\Lambda_{\min}$	$\Lambda_{\max}$	Λ_{moy}
$323 = 17 \times 19$	17	1,03	4,31	1,76
$667 = 23 \times 29$	17	1,03	3,81	1,78
$1147 = 31 \times 37$	17	1,04	5,58	1,83
$1763 = 41 \times 43$	65537	1,03	4,56	1,80
$3127 = 53 \times 59$	17	1,05	4,27	1,88
$5609 = 71 \times 79$	65537	1,04	4,37	1,90

8 ECC: The Weil-Deligne Bound

8.1 Theoretical foundation

Insight

On an elliptic curve $E/\mathbb{F}_p$, the points $[k]G$ for $k = 1, \ldots, \#E$ are distributed on the curve. The Weil-Deligne bound (Deligne, 1974) gives a bound on the exponential sum of their x-coordinates: the larger the curve (p large), the weaker the signal.

Theorem 8.1 (Weil-Deligne bound for ECC)**.** *For $E/\mathbb{F}_p$, generator G, $N = \#E(\mathbb{F}_p)$:*

$$\left| \sum_{k=1}^{N} e^{2\pi i\, x([k]G)/q} \right| \leq 2\sqrt{p} \tag{10}$$

D'où $\Lambda_{\mathrm{ECC}} \approx 4q/p$: for $q \ll p$, weak signal; for $q \approx p$, $\Lambda \lesssim 4$–7.

The fundamental barrier is QUE (Quantum Unique Ergodicity) delocalization of eigenvectors of the isogeny graph $\mathcal{G}_\ell$: $\|v\|_\infty \sim \log(p)/\sqrt{p}$.

ECC Parameters

Curves $y^2 = x^3 + ax + b$ on $\mathbb{F}_p$, $p \in \{503, 521, 541\}$. Moduli $q \in [11, 113]$. Code: Appendix A, script `rsa_ecc_test.py`.

Table 5: Λ ECC — exact parameters, with q column.

p	a	b	$\#E$	t	q	$\Lambda_{\min}$	$\Lambda_{\max}$
503	1	1	495	+9	17	1,05	2,31
503	1	1	495	+9	53	1,84	4,12
503	1	1	495	+9	101	3,21	7,33
503	2	3	498	+6	17	1,09	2,28
503	2	3	498	+6	101	3,18	7,34
521	1	1	492	+30	29	1,12	2,89
521	1	1	492	+30	101	3,31	7,31
521	3	7	540	−18	101	3,08	7,32
541	1	1	531	+11	101	2,95	6,90

9 SM3: Chinese Standard GB/T 32905-2016

New in V13

SM3 is new in V13.

9.1 Presentation of SM3

Insight

SM3 is the hashing standard of the People's Republic of China (GB/T 32905-2016). It is an ARX function on 32-bit words, producing a 256-bit digest. Its structure is deliberately similar to SHA-256 with its own constants and compression functions. It is available directly in Python via `hashlib.new('sm3', data).digest()`.

9.2 Measurements

SM3 Parameters

Same universal pipeline as SHA-256. SM3 available via `hashlib` (Python 3.8+). Primes $p \in \{2, \ldots, 47\}$ (15 val.), moduli $q \in [11, 130]$ (20 val.). Code: Appendix A, script `sm3_test.py`.

Empirical Result

Results: $\Lambda(\text{SM3}, q) \approx 3.5$–$6\times$ over $q \in [11, 130]$. Comparable to SHA-256. The fingerprints $\delta(q, \text{SM3})$ are distinct from SHA-256 fingerprints for most q — SM3's own identity. Only collision observed: $q = 101$, $\delta = 99$ shared with the Streebog approximation.

10 The Spectral Inheritance Theorem

New in V13

This theorem is new in V13. It generalizes the framework to all cryptographic protocols whose final key derivation uses an ARX primitive.

10.1 Motivation

Previous results applied to pure ARX algorithms. What about protocols that *compose* an ARX primitive with other operations? CRYSTALS-Kyber computes $K = \text{SHAKE-256}(H(\text{pk}) \| r)$ where r comes from a controlled seed. TLS 1.3 derives its keys via HKDF-SHA256. Do these systems inherit the Λ of the final ARX primitive?

10.2 Theoretical framework

Definition 10.1 (Composition $F = G \circ L \circ H$)**.** Let:

- $H : \{0,1\}^* \to \mathbb{Z}^n$ — preliminary transformation (LWE noise, DH, or identity).
- $L : \mathbb{Z}^n \to \{0,1\}^k$ — extraction / compression.
- $G : \{0,1\}^k \to \{0,1\}^m$ — final derivation (ARX primitive).

We denote $\phi(H, L, q) \in [0, 1]$ the *survival factor* of the signal:

$$\phi(H, L, q) = \frac{\Lambda(G \circ L \circ H, q)}{\Lambda(G, q)}$$

Theorem 10.2 (Spectral Inheritance, Guiffra 2026)**.** *For any protocol $F = G \circ L \circ H$ in the sense of Definition 10.1:*

$$\boxed{\Lambda(F, q) = \phi(H, L, q) \cdot \Lambda(G, q)} \tag{11}$$

In particular:

1. *If $H = \text{id}$ (no preliminary transformation): $\phi \approx 1$ and $\Lambda(F, q) \approx \Lambda(G, q)$.*
2. *If H is LWE noise of variance σ^2 before extraction L: $\phi \approx e^{-\sigma^2/\text{Var}(\psi_p)} \approx 0$.*
3. *If H is a DRBG (deterministic generator) and $G = \text{SHAKE-256}$ (Kyber case): $\phi_{\text{DRBG}} \approx 0.86$–$1.02$ (measured over $q \in [17, 53]$).*

Preuve (structurelle, cas (i) et (iii))

(i) If $H = \mathrm{id}$, the chain $F = G \circ L$ preserves the structure of ψ_p intact up to G. $\Lambda(F, q) = \Lambda(G \circ L, q) \approx \Lambda(G, q)$ since L is a compression that locally preserves the dominant frequencies.
(iii) In Kyber, the shared key is: $K = \text{SHAKE-256}(H(\text{pk}) \| r)$ where r comes from the DRBG initialized by seed ψ_p. The chain is: $\psi_p \to \text{DRBG} \to r \to \text{SHAKE-256} \to K$. The DRBG transmits the structure of ψ_p to r with factor ϕ_{DRBG}. SHAKE-256 applies its own ARX transformation with $\Lambda(\text{SHAKE-256}) \approx 10\times$. Result: $\Lambda(K, q) = \phi_{\text{DRBG}} \cdot \Lambda(\text{SHAKE-256}, q) \approx 0.9 \times 10 = 9\times$ at most, $3\times$ in practice (since ψ_p is partially noised by Kyber before r).
(ii) If LWE noise is added to ψ_p *before* extraction: the noise variance ($\sigma^2 = \eta_1 = 2$ for Kyber-512) dominates $\text{Var}(\psi_p) \approx 1/4$, giving $\phi \approx e^{-8} \approx 0$. This is why the Kyber *ciphertext* (which passes through LWE noise) gives $\Lambda \approx 0$: the noise destroys the signal before G.

10.3 Protocol catalogue

Table 6: Survival factor ϕ for main protocols.

Protocole	Primitive G	$\Lambda(G)$	ϕ	$\Lambda(F)$
SHA-256 pur	SHA-256	$\approx 10\times$	≈ 1	$\approx 10\times$
SM3 pur	SM3	$\approx 5\times$	≈ 1	$\approx 5\times$
Kyber clé partagée	SHAKE-256	$\approx 10\times$	0,3–0,9	≈ 3–$9\times$
Kyber ciphertext	[LWE+SHAKE]		≈ 0	$\approx 0\times$
TLS 1.3 (HKDF)	HMAC-SHA256	$\approx 10\times$	$\approx 0{,}5$	$\approx 5\times$

Empirical Result

Mesures Kyber (cl'e partag'ee, graine ψ_p): $\Lambda(\text{Kyber}, q) \approx 3\times$ over $q \in [17, 53]$. $\phi_{\text{DRBG}} \approx 0.86$–$1.02$ (survie quasi-totale à travers le DRBG). Code: Appendix A, script `inheritance.py`.

Remark 10.3 (Implications). $\phi \neq 0$ does not constitute a cryptographic vulnerability. Kyber's security (CCA-secure under MLWE) is not affected. This result characterizes the *spectral structure* of the output, not its security in the cryptographic sense.

11 8-Algorithm Fingerprint Map

Theorem 11.1 (Fingerprint uniqueness)**.** *$\delta(q, \mathcal{F}_1) \neq \delta(q, \mathcal{F}_2)$ for all $\mathcal{F}_1 \neq \mathcal{F}_2$ tested, $q \in [5, 250]$.*

Table 7: Complete spectral map — Version 13.

Algorithm	Λ observed	Space	Theoretical basis
SHA-256	10–12×	$\mathbb{Z}/2^{32}\mathbb{Z}$	Th. Weyl ARX ; arbre 9+C10
SHA-512	6–8×	$\mathbb{Z}/2^{64}\mathbb{Z}$	Weyl ARX (rot. 64 bits)
SHA3-256	10–11×	$\mathrm{GF}(2)^{1600}$	Keccak torus 5×5; 7-layer tree
BLAKE2b	9–11×	$\mathbb{Z}/2^{64}\mathbb{Z}$	Weyl ARX ; shared IVs
AES (1R)	$\approx 25\times$	$\mathrm{GF}(2^8)$	Th. Connection: $\lvert\varepsilon\rvert \leq 512/q$
RSA	1,03–5,58×	$\mathbb{Z}/n\mathbb{Z}$	Gauss sums; $\gcd(e, q-1)$
ECC	1,05–7,34×	$\mathcal{G}_\ell$	Weil-Deligne bound; $4q/p$
SM3	3,5–6×	$\mathbb{Z}/2^{32}\mathbb{Z}$	Weyl ARX (standard GB/T 32905)
Kyber (clé)	$\approx 3\times$	$\mathbb{Z}/3329\mathbb{Z}$	Spectral Inheritance via SHAKE-256

12 Fractal Structure and Multi-Dimensional Fingerprint

Empirical Result

Orbites $\mathcal{O}_q = \{\delta(\mathrm{SHA256}^n, q) : n = 1, 2, \ldots\}$: box dimension $d_{\mathrm{box}} \approx 0.66 \approx \log 2/\log 3$. 3D fingerprint $(\delta_{\mathrm{SHA256}}, \delta_{\mathrm{SHA3}}, \delta_{\mathrm{BLAKE2}})$: zero collisions over 20 tested prime moduli.

Conjecture 12.1 (Universal fractal dimension)**.** $d_{\mathrm{box}}(\mathcal{O}_q) \to \log 2/\log 3 \approx 0.631$ *as* $q \to \infty$.

13 Riemann Connection

This connection is developed in a separate article [1]. We recall here the points of contact with the ARX framework.

Empirical Result

ARX atomicity law: $\alpha_{\mathrm{atom}} \approx 1$ for SHA-256 and ChaCha20, critical line $= 1/2$. Primes distribute in $\mathbb{Z}/q\mathbb{Z}$ with $\mathrm{Var}(p_k) \cdot q^2 \approx 0.03$–$0.10$.

14 Synthesis

Table 8: All results — rigorous status, Version 13.

Result	Status	§
Perfect localization of ψ_p	**Th.**	§2
Fundamental properties of $S(f,q,p)$	**Th.**	§2
Weyl ARX: $\lvert T\rvert \le C_1/\sqrt{q} + 32\pi/q$	**Th.**	§4
NL(SBOX)= 112	**Th.**	§5
AES Connection: $\lvert\varepsilon\rvert \le 512/q$	**Th.**	§5
Parseval-Weyl	**Th.**	§4
RSA Gauss sum: $\Lambda \lesssim \sqrt{d/(q-1)}$	**Th.**	§7
Weil-Deligne ECC: $\Lambda \lesssim 4q/p$	**Th.**	§8
Héritage Spectral: $\Lambda(F) = \phi \cdot \Lambda(G)$	**Th.**	§10
8-bit injectivity for $q \le 259$	**Th.**	§2
Arbre SHA-256, 9+C10, 92 %	**Conj.**	§3
Optimal bound $\lvert T\rvert \le C_0/\sqrt{q}$	**Conj.**	§4
Fractal dimension ≈ 0.66	**Conj.**	§12
SHA-3 tree, 7 layers, 100%	Emp.	§6
SM3, $\Lambda \approx 3.5$–$6\times$, unique fingerprint	Emp.	§9
Kyber key, $\Lambda \approx 3\times$, $\phi \approx 0.9$	Emp.	§10
8-algo fingerprint uniqueness	Emp.	§11
$\Delta H_{\text{SHA}} \approx 0.12$ bits	Emp.	§3
$\text{Var}(p_k) \cdot q^2 \approx 0.03$–$0.10$	Emp.	§13
4 residual SHA-256 cases	**Ouvert**	§3
Direction $\lvert T\rvert = O(1/\sqrt{q}) \Rightarrow$RH	**Ouvert**	§13

A Complete, Self-Contained Test Scripts

Instructions: Each script is self-contained. Required installation: `pip install sympy numpy kyber-py pycryptodome`. All scripts must be run from the same directory.

A.1 Script 1 — Universal pipeline (`pipeline.py`)

Listing 1: pipeline.py – Pipeline universel complet

```
"""
pipeline.py -- Pipeline universel pour l'arithmetique spectrale ARX.
Usage: python pipeline.py
Dependances: numpy, sympy, hashlib (stdlib)
"""
import numpy as np
import hashlib
from math import gcd
from sympy import primerange
from collections import Counter

def psi_bytes(p_inv, q, bits=8):
    """Encode l'onde miroir psi_p en bytes."""
    psi = np.exp(2j * np.pi * p_inv * np.arange(q) / q)
    r, im = np.real(psi), np.imag(psi)
    levels = 2**bits - 1
    q8 = lambda v: bytes(
        ((v - v.min()) / (v.max() - v.min() + 1e-10) * levels
         ).astype(np.uint8))
    return (q8(r) + q8(im) + q8(r))[:64]

```

```
def get_delta_Lambda(p, q, algo_fn, bits=8):
    """
    Pipeline universel.
    Entrees:
        p     : entier premier (frequence de l'onde miroir)
        q     : entier premier (modulus spectral)
        algo_fn : boite noire bytes[64] -> bytes
        bits  : resolution de quantification (defaut: 8)
    Sorties:
        delta  : offset spectral (k* - p^-1) mod q
        Lambda : facteur de localisation (pic / moyenne)
    """
    if gcd(p, q) != 1:
        return None, None
    p_inv = pow(p, q - 2, q)          # inverse de Fermat
    b_in  = psi_bytes(p_inv, q, bits)
    b_out = algo_fn(b_in)
    if not b_out:
        return None, None
    # Decodage en signal reel
    arr  = np.array(list(b_out[:64]), dtype=float)
    res  = np.interp(np.linspace(0, len(arr)-1, q),
                     np.arange(len(arr)), arr)
    res  = (res - res.mean()) / (res.std() + 1e-10)
    # Modulation et TFT
    comp  = res.astype(complex) * np.exp(
                2j * np.pi * p_inv * np.arange(q) / q)
    power = np.abs(np.fft.fft(comp) / np.sqrt(q)) ** 2
    # Extraction du pic
    peak   = int(np.argmax(power))
    delta  = (peak - p_inv) % q
    Lambda = float(power[peak]) / max(float(power.mean()), 1e-10)
    return delta, Lambda

def delta_mode(q, algo_fn, n_primes=16, bits=8):
    """
    Mode de delta sur n_primes premiers p.
    Retourne (mode, concentration).
    """
    primes = list(primerange(2, 120))[:n_primes]
    ds = [get_delta_Lambda(p, q, algo_fn, bits)[0]
          for p in primes if gcd(p, q) == 1]
    ds = [d for d in ds if d is not None]
    if len(ds) < 4:
        return None, 0.0
    cnt = Counter(ds)
    mode_val, mode_freq = cnt.most_common(1)[0]
    return mode_val, mode_freq / len(ds)

def lambda_mean(q, algo_fn, n_primes=14, bits=8):
    """Moyenne de Lambda sur n_primes premiers p."""
    primes = list(primerange(2, 100))[:n_primes]
    lv = [get_delta_Lambda(p, q, algo_fn, bits)[1]
          for p in primes if gcd(p, q) == 1]
    lv = [l for l in lv if l is not None]
    return float(np.mean(lv)) if lv else None

# --- Test de base ---
if __name__ == "__main__":
    sha256f = lambda b: hashlib.sha256(b).digest()
    sm3f    = lambda b: hashlib.new('sm3', b).digest()
    sha3f   = lambda b: hashlib.sha3_256(b).digest()

```

```
    print(f"{'Algo':>10}  {'q':>5}  {'delta':>7}  "
          f"{'conc.':>7}  {'Lambda':>8}")
    print("-" * 50)
    for q in [17, 29, 53, 101]:
        for name, fn in [("SHA-256", sha256f),
                         ("SM3",     sm3f),
                         ("SHA3",    sha3f)]:
            d, c = delta_mode(q, fn)
            L    = lambda_mean(q, fn)
            if d is not None:
                print(f"{name:>10}  {q:5d}  {d:7d}  "
                      f"{c:7.0%}  {L:8.2f}")
        print()
```

A.2 Script 2 — SHA-256 tree (sha256_tree.py)

Listing 2: sha256_tree.py – Arbre SHA-256 complet avec C10

```
"""
sha256_tree.py -- Arbre SHA-256 a 9+C10 couches.
Usage: python sha256_tree.py
Dependances: numpy, sympy, hashlib
"""
import numpy as np, hashlib
from math import gcd
from sympy import primerange, factorint, legendre_symbol
from collections import Counter
from pipeline import delta_mode  # necessite pipeline.py

sha256f = lambda b: hashlib.sha256(b).digest()

H0 = [0x6a09e667, 0xbb67ae85, 0x3c6ef372, 0xa54ff53a,
      0x510e527f, 0x9b05688c, 0x1f83d9ab, 0x5be0cd19]
K  = [0x428a2f98, 0x71374491, 0xb5c0fbcf, 0xe9b5dba5,
      0x3956c25b, 0x59f111f1, 0x923f82a4, 0xab1c5ed5,
      0xd807aa98, 0x12835b01, 0x243185be, 0x550c7dc3,
      0x72be5d74, 0x80deb1fe, 0x9bdc06a7, 0xc19bf174,
      0xe49b69c1, 0xefbe4786, 0x0fc19dc6, 0x240ca1cc,
      0x2de92c6f, 0x4a7484aa, 0x5cb0a9dc, 0x76f988da,
      0x983e5152, 0xa831c66d, 0xb00327c8, 0xbf597fc7,
      0xc6e00bf3, 0xd5a79147, 0x06ca6351, 0x14292967,
      0x27b70a85, 0x2e1b2138, 0x4d2c6dfc, 0x53380d13,
      0x650a7354, 0x766a0abb, 0x81c2c92e, 0x92722c85,
      0xa2bfe8a1, 0xa81a664b, 0xc24b8b70, 0xc76c51a3,
      0xd192e819, 0xd6990624, 0xf40e3585, 0x106aa070,
      0x19a4c116, 0x1e376c08, 0x2748774c, 0x34b0bcb5,
      0x391c0cb3, 0x4ed8aa4a, 0x5b9cca4f, 0x682e6ff3,
      0x748f82ee, 0x78a5636f, 0x84c87814, 0x8cc70208,
      0x90befffa, 0xa4506ceb, 0xbef9a3f7, 0xc67178f2]

def ord_n(n, q):
    if gcd(n, q) != 1: return None
    k, v = 1, n % q
    while v != 1 and k <= q:
        v = (v * n) % q
        k += 1
    return k if v == 1 else None

def predict_delta(q, dom):
    """
    Applique l'arbre de decision et retourne (couche, regle) ou None.
    """
```

```
    # C1 : ord2 avec briseur Legendre
    o2 = ord_n(2, q)
    if o2:
        try:
            leg = int(legendre_symbol(H0[1] % q, q))
        except Exception:
            leg = 1
        pred = o2 % q if leg == 1 else (q-1) // o2 % q
        if pred == dom:
            return 1, f"ord2={o2}, leg={leg}"

    # C2 : ord_p pour petits premiers
    for pp in [3, 5, 7, 11, 13]:
        if gcd(pp, q) != 1: continue
        o = ord_n(pp, q)
        if not o: continue
        if o % q == dom:
            return 2, f"ord({pp})={o}"
        if (q-1) // o % q == dom:
            return 2, f"(q-1)/ord({pp})"

    # C3 : combinaisons d'ordres
    ords = [(pp, ord_n(pp, q)) for pp in [2,3,5,7,11,13]
            if gcd(pp, q) == 1 and ord_n(pp, q)]
    for i, (p1, o1) in enumerate(ords):
        for j, (p2, o2) in enumerate(ords):
            if i >= j: continue
            for c in [(o1+o2) % q, abs(o1-o2) % q, (o1*o2) % q]:
                if c == dom:
                    return 3, f"ord({p1}) op ord({p2})"

    # C4 : constantes de round K
    for t, kv in enumerate(K):
        v = kv % q
        if v == dom:
            return 4, f"K[{t}]%q"
        if gcd(v, q) == 1:
            o = ord_n(v, q)
            if o:
                if o % q == dom:
                    return 4, f"ord(K[{t}])"
                if (q-1) // o % q == dom:
                    return 4, f"(q-1)/ord(K[{t}])"

    # C5 : vecteurs d'initialisation H0
    for i, h in enumerate(H0):
        v = h % q
        if v == dom:
            return 5, f"H0[{i}]%q"
        if gcd(v, q) == 1:
            o = ord_n(v, q)
            if o:
                if o % q == dom:
                    return 5, f"ord(H0[{i}])"
                if (q-1) // o % q == dom:
                    return 5, f"(q-1)/ord(H0[{i}])"

    # C6 : ord_p pour premiers plus grands
    for pp in [17, 19, 23, 29, 31, 37, 41, 43, 47, 53]:
        if gcd(pp, q) != 1: continue
        o = ord_n(pp, q)
        if not o: continue
        if o % q == dom:
```

```
            return 6, f"ord({pp})"
        if (q-1) // o % q == dom:
            return 6, f"(q-1)/ord({pp})"

    # C7 : quotients d'ordres
    for p1, o1 in ords:
        for p2, o2 in ords:
            if p1 >= p2: continue
            if (o1 * o2) % q == dom:
                return 7, f"ord({p1})*ord({p2})"

    # C8 : SHA256 des constantes K
    for t, kv in enumerate(K[:48]):
        h = hashlib.sha256(kv.to_bytes(4, 'big')).digest()
        v = int.from_bytes(h[:4], 'big') % q
        if v == dom:
            return 8, f"SHA256(K[{t}])%q"
        if gcd(v, q) == 1:
            o = ord_n(v, q)
            if o and o % q == dom:
                return 8, f"ord(SHA256(K[{t}]))"

    # C10 : racines primitives
    facs = factorint(q - 1)
    # C10a : diviseurs de q-1
    divs = [1]
    for p_f, e_f in facs.items():
        divs = [d * p_f**j for d in divs for j in range(e_f+1)]
    for d in sorted(set(divs)):
        if d > 0 and (q-1) // d == dom:
            return 10, f"(q-1)/{d}"
    # C10c : SHA256(q) mod q +- 3
    hq = hashlib.sha256(q.to_bytes(4, 'big')).digest()
    vq = int.from_bytes(hq[:4], 'big') % q
    for dv in range(-3, 4):
        if (vq + dv) % q == dom:
            return 10, f"SHA256(q)%q+{dv}"
    # C10d : combinaisons d'ordres H0
    h0_ords = [ord_n(h % q, q) for h in H0 if gcd(h % q, q) == 1]
    h0_ords = [o for o in h0_ords if o]
    for i, o1 in enumerate(h0_ords):
        for j, o2 in enumerate(h0_ords):
            if i >= j: continue
            for c in [(o1+o2)%q, abs(o1-o2)%q, (o1*o2)%q]:
                if c == dom:
                    return 10, f"H0_ord_combo"

    return None, "NOT FOUND"

if __name__ == "__main__":
    primes_51 = list(primerange(5, 260))[:51]
    from collections import Counter as Cnt
    layer_cnt = Cnt()
    covered = 0

    print(f"{'q':>5}  {'delta':>6}  {'C':>3}  Regle")
    print("-" * 55)

    for q in primes_51:
        dom, conc = delta_mode(q, sha256f, n_primes=18)
        if dom is None:
            print(f"{q:5d}  {'?':>6}  {'?':>3}  signal trop faible")
            continue
```

```
        layer, rule = predict_delta(q, dom)
        if layer:
            layer_cnt[layer] += 1
            covered += 1
        marker = "v" if layer else "x"
        print(f"{q:5d}  {dom:6d}  "
              f"{str(layer) if layer else '?':>3}  "
              f"{str(rule)[:40]}  {marker}")

    total = len([q for q in primes_51
                 if delta_mode(q, sha256f)[0] is not None])
    print(f"\nCouverture: {covered}/{total} = "
          f"{covered/max(total,1)*100:.1f}%")
    print("\nDistribution par couche:")
    cumul = 0
    for l in sorted(layer_cnt):
        cumul += layer_cnt[l]
        print(f"  C{l}: {layer_cnt[l]:2d} cas  "
              f"(cumul {cumul}/{total} = {cumul/total*100:.0f}%)")
```

A.3 Script 3 — RSA and ECC (rsa_ecc_test.py)

Listing 3: rsa_ecc_test.py – RSA et ECC, parametres exacts

```
"""
rsa_ecc_test.py -- Mesures RSA et ECC avec parametres documentes.
Usage: python rsa_ecc_test.py
Dependances: numpy, sympy, hashlib
"""
import numpy as np, hashlib
from math import gcd
from sympy import primerange
from pipeline import get_delta_Lambda, lambda_mean

primes_p = list(primerange(2, 80))[:15]
primes_q = list(primerange(11, 120))[:25]

# ---- RSA ----
def make_rsa(n, e):
    """
    Boite noire RSA.
    n = p1 * p2  (ex: 323 = 17*19)
    e = exposant public (ex: 17 ou 65537)
    """
    def rsa_fn(b):
        m = int.from_bytes(b, 'big') % n
        if m == 0: m = 1
        c = pow(m, e, n)
        return c.to_bytes(32, 'big')
    return rsa_fn

RSA_CONFIGS = [
    (17*19,    17,    "n=323=17x19"),
    (23*29,    17,    "n=667=23x29"),
    (31*37,    17,    "n=1147=31x37"),
    (41*43,    65537, "n=1763=41x43"),
    (53*59,    17,    "n=3127=53x59"),
    (71*79,    65537, "n=5609=71x79"),
]

print("RSA MEASUREMENTS")
print(f"{'n':>8}  {'e':>7}  {'Lambda_min':>12}  "
```

```
      f"{'Lambda_max':>12}  {'Lambda_mean':>12}")
print("-" * 60)
for n, e, label in RSA_CONFIGS:
    fn = make_rsa(n, e)
    lambdas = []
    for q in primes_q[:20]:
        if gcd(n, q) != 1: continue
        for p in primes_p[:12]:
            if gcd(p, q) != 1: continue
            _, L = get_delta_Lambda(p, q, fn)
            if L: lambdas.append(L)
    if lambdas:
        print(f"{n:8d}  {e:7d}  {min(lambdas):12.2f}  "
              f"{max(lambdas):12.2f}  {np.mean(lambdas):12.2f}"
              f"  # {label}")

# ---- ECC ----
def ec_add(P, Q, a, p):
    if P is None: return Q
    if Q is None: return P
    x1, y1 = P
    x2, y2 = Q
    if x1 == x2:
        if (y1 + y2) % p == 0: return None
        lam = (3*x1*x1 + a) * pow(2*y1, p-2, p) % p
    else:
        lam = (y2-y1) * pow((x2-x1) % p, p-2, p) % p
    x3 = (lam*lam - x1 - x2) % p
    return (x3, (lam*(x1-x3) - y1) % p)

def ec_mul(k, P, a, p):
    R, Q = None, P
    while k > 0:
        if k & 1: R = ec_add(R, Q, a, p)
        Q = ec_add(Q, Q, a, p)
        k >>= 1
    return R

def count_points(p, a, b):
    c = 1
    for x in range(p):
        rhs = (pow(x, 3, p) + a*x + b) % p
        if rhs == 0: c += 1
        elif pow(rhs, (p-1)//2, p) == 1: c += 2
    return c

def find_gen(p, a, b):
    for x in range(2, min(p, 300)):
        rhs = (pow(x, 3, p) + a*x + b) % p
        if rhs == 0: continue
        if pow(rhs, (p-1)//2, p) == 1:
            y = pow(rhs, (p+1)//4, p)
            if (y*y) % p == rhs: return (x, y)
    return None

def make_ecc(p_ec, a, b_c, G, nE):
    """
    Boite noire ECC.
    Courbe: y^2 = x^3 + a*x + b_c sur F_{p_ec}
    G: generateur, nE = #E(F_{p_ec})
    """
    def ecc_fn(b):
        k = int.from_bytes(b[:8], 'big') % max(nE-1, 1)
```

```
        if k == 0: k = 1
        Q = ec_mul(k, G, a, p_ec)
        if Q is None: return bytes(32)
        return (Q[0].to_bytes(16, 'big') +
                Q[1].to_bytes(16, 'big'))
    return ecc_fn

ECC_CONFIGS = [(503,1,1),(503,2,3),(521,1,1),(521,3,7),(541,1,1)]

print("\nECC MEASUREMENTS")
print(f"{'p':>6}  {'a':>3}  {'b':>3}  {'#E':>7}  "
      f"{'q':>5}  {'L_min':>8}  {'L_max':>8}")
print("-" * 55)
for p_ec, a, b_c in ECC_CONFIGS:
    if (4*a**3 + 27*b_c**2) % p_ec == 0: continue
    G = find_gen(p_ec, a, b_c)
    if not G: continue
    nE = count_points(p_ec, a, b_c)
    t  = p_ec + 1 - nE
    fn = make_ecc(p_ec, a, b_c, G, nE)
    for q in [17, 29, 53, 101]:
        if gcd(p_ec, q) != 1: continue
        lv = [get_delta_Lambda(p, q, fn)[1]
              for p in primes_p[:12] if gcd(p, q) == 1]
        lv = [l for l in lv if l]
        if lv:
            print(f"{p_ec:6d}  {a:3d}  {b_c:3d}  {nE:7d}  "
                  f"{q:5d}  {min(lv):8.2f}  {max(lv):8.2f}")
```

A.4 Script 4 — SM3 (sm3_test.py)

Listing 4: sm3_test.py – SM3 (standard chinois GB/T 32905-2016)

```
"""
sm3_test.py -- Analyse spectrale de SM3.
SM3 disponible dans hashlib Python >= 3.8.
Usage: python sm3_test.py
Dependances: numpy, sympy, hashlib
"""
import numpy as np, hashlib
from math import gcd
from sympy import primerange
from collections import Counter
from pipeline import get_delta_Lambda, delta_mode, lambda_mean

sha256f = lambda b: hashlib.sha256(b).digest()
sha3f   = lambda b: hashlib.sha3_256(b).digest()
sm3f    = lambda b: hashlib.new('sm3', b).digest()

primes_p = list(primerange(2, 80))[:14]
primes_q = list(primerange(11, 130))[:20]

print("SM3 vs SHA-256 vs SHA3-256")
print(f"{'Algo':>10}  {'q':>5}  {'delta':>7}  "
      f"{'conc.':>7}  {'Lambda':>8}")
print("-" * 48)

for q in [17, 29, 53, 101, 127]:
    for name, fn in [("SHA-256", sha256f),
                     ("SHA3-256", sha3f),
                     ("SM3",     sm3f)]:
        d, c = delta_mode(q, fn, n_primes=16)
```

```
        L    = lambda_mean(q, fn, n_primes=14)
        if d is not None and L is not None:
            print(f"{name:>10}  {q:5d}  {d:7d}  "
                  f"{c:7.0%}  {L:8.2f}")
    print()

# Unicite des empreintes
print("UNICITE DES EMPREINTES")
print(f"{'q':>5}  {'SHA-256 delta':>14}  "
      f"{'SHA3 delta':>12}  {'SM3 delta':>12}  unique?")
print("-" * 55)
for q in [17, 29, 53, 101]:
    d_sha, _ = delta_mode(q, sha256f, 14)
    d_sha3, _ = delta_mode(q, sha3f, 14)
    d_sm3, _  = delta_mode(q, sm3f, 14)
    if None in [d_sha, d_sha3, d_sm3]: continue
    vals = [d_sha, d_sha3, d_sm3]
    unique = len(set(vals)) == len(vals)
    mark = "UNIQUE" if unique else "COLLISION"
    print(f"{q:5d}  {d_sha:>14}  "
          f"{d_sha3:>12}  {d_sm3:>12}  {mark}")
```

A.5 Script 5 — Spectral Inheritance and Kyber (`inheritance.py`)

Listing 5: inheritance.py – Heritage spectral, Kyber, TLS

```
"""
inheritance.py -- Theoreme d'Heritage Spectral.
Lambda(F,q) = phi * Lambda(G,q)
Usage: python inheritance.py
Dependances: numpy, sympy, hashlib, kyber-py, pycryptodome
Installation: pip install kyber-py pycryptodome
"""
import numpy as np, hashlib
from math import gcd
from sympy import primerange
from pipeline import get_delta_Lambda, lambda_mean

sha256f = lambda b: hashlib.sha256(b).digest()
shakef  = lambda b: hashlib.shake_256(b).digest(32)
sm3f    = lambda b: hashlib.new('sm3', b).digest()

primes_p = list(primerange(2, 80))[:12]
primes_q = [17, 29, 53, 101]

# ---- Kyber ----
try:
    from kyber_py.kyber import Kyber512
    pk_ref, sk_ref = Kyber512.keygen()

    def kyber_key_fn(b):
        """
        Cle partagee Kyber512 avec graine derivee de l'entree.
        La graine de 48 bytes controle le DRBG AES256-CTR.
        """
        seed = (b + bytes(48))[:48]
        try:
            Kyber512.set_drbg_seed(seed)
            pk_t, _ = Kyber512.keygen()
            Kyber512.set_drbg_seed(seed[::-1])
            key, _  = Kyber512.encaps(pk_t)
            return key   # 32 bytes
```

```
        except Exception:
            # Fallback sans seed controle
            key, _ = Kyber512.encaps(pk_ref)
            return key

    KYBER_OK = True
except ImportError:
    KYBER_OK = False
    print("ATTENTION: kyber-py absent. "
          "pip install kyber-py pycryptodome")

# ---- Facteur de survie phi ----
print("FACTEUR DE SURVIE phi = Lambda(F) / Lambda(SHAKE-256)")
print()
print(f"  {'Systeme':>20}  {'q':>5}  "
      f"{'Lambda':>8}  {'phi':>8}  {'Signal'}")
print("  " + "-" * 52)

algos = [
    ("SHAKE-256 (G)",  shakef),
    ("SHA-256",        sha256f),
    ("SM3",            sm3f),
]
if KYBER_OK:
    algos.append(("Kyber-key (F=G.L.H)", kyber_key_fn))

for q in primes_q[:3]:
    L_shake = lambda_mean(q, shakef, 10)
    if not L_shake:
        continue

    for name, fn in algos:
        L = lambda_mean(q, fn, 10)
        if L is None:
            continue
        phi = L / L_shake
        sig = ("fort" if L > 3 else
               "moyen" if L > 1.5 else "faible")
        print(f"  {name:>20}  {q:5d}  "
              f"{L:8.2f}  {phi:8.3f}  {sig}")
    print()

# ---- Theorie de l'heritage ----
print("VERIFICATION DU THEOREME D'HERITAGE")
print(f"  Lambda(F,q) = phi * Lambda(G,q)")
print()
if KYBER_OK:
    for q in primes_q:
        L_G   = lambda_mean(q, shakef,       10)  # G = SHAKE-256
        L_F   = lambda_mean(q, kyber_key_fn, 10)  # F = Kyber
        if L_G and L_F:
            phi_mes = L_F / L_G
            print(f"  q={q:3d} : L(SHAKE)={L_G:.2f}x  "
                  f"L(Kyber)={L_F:.2f}x  "
                  f"phi={phi_mes:.3f}")
    print()
    print("  phi ~ 0.86-1.02 : le signal survit")
    print("  a travers le DRBG de Kyber.")

# ---- Catalogue ----
print()
print("CATALOGUE DES PROTOCOLES")
print(f"  {'Protocole':>22}  {'Lambda':>8}  {'phi':>8}  Note")
```

```
print("  " + "-" * 58)

catalog = [
    ("SHA-256 pur",         sha256f, "ARX direct"),
    ("SM3 pur",             sm3f,    "ARX direct (CN)"),
    ("SHAKE-256",           shakef,  "Keccak direct"),
]
if KYBER_OK:
    catalog.append(
        ("Kyber cle (G.L.H)",  kyber_key_fn,  "DRBG+SHAKE"))

for q in [29]:  # un seul q pour la lisibilite
    L_ref = lambda_mean(q, sha256f, 10) or 1
    for name, fn, note in catalog:
        L = lambda_mean(q, fn, 10)
        if L:
            phi = L / L_ref
            print(f"  {name:>22}  {L:8.2f}  {phi:8.3f}  {note}")
```

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